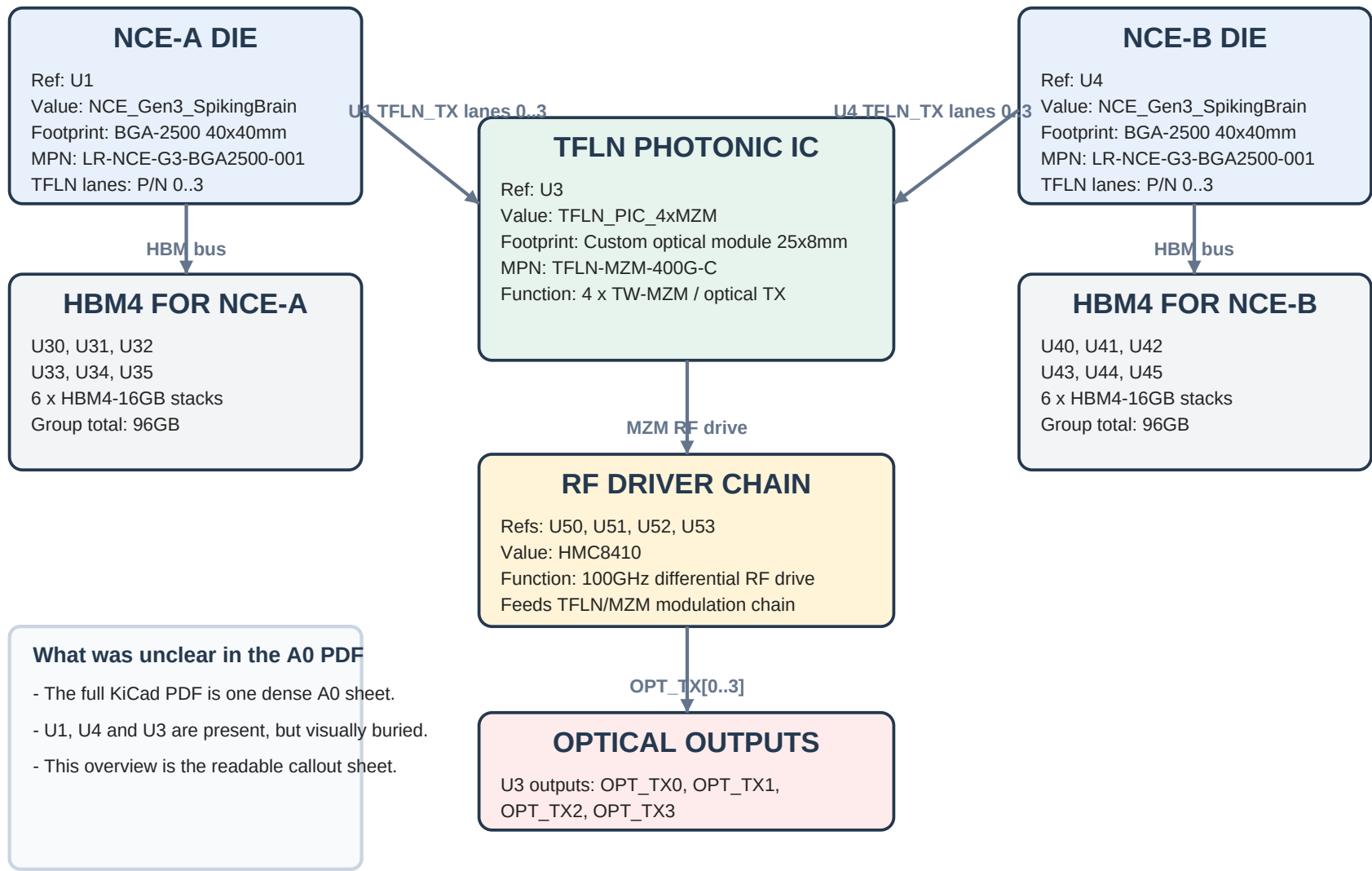


TFLN_AI_NODE_X2 - Clear Critical Schematic Overview

Dual NCE dies and TFLN photonic IC are present in the native KiCad schematic; this page makes them explicit for manufacturer review.

Source: TFLN_AI_NODE_X2_SERVER_CHASSIS_IO.kicad_sch | Status: Engineering ECO required / release pending



Critical Component Cross-Reference

Use this page to verify that dual NCE dies and the TFLN photonic IC are represented in the schematic/BOM set.

Ref	Block	Value / MPN	Footprint	Role
U1	NCE-A die	NCE_Gen3_SpikingBrain LR-NCE-G3-BGA2500-001	LightRail:BGA-2500_40x40mm_P0.8mm	Neural compute engine A. Drives TFLN TX lane group P/N 0..3 and HBM group U30-U35.
U4	NCE-B die	NCE_Gen3_SpikingBrain LR-NCE-G3-BGA2500-001	LightRail:BGA-2500_40x40mm_P0.8mm	Neural compute engine B. Drives TFLN TX lane group P/N 0..3 and HBM group U40-U45.
U3	TFLN photonic IC	TFLN_PIC_4xMZM TFLN-MZM-400G-C	LightRail:Custom_Optical_Module_2 5x8mm	TFLN/LiNbO3 photonic IC, 4 x TW-MZM, optical TX outputs OPT_TX0..OPT_TX3.
U50-U53	RF driver chain	HMC8410 HMC8410LP2FE	QFN-32-1EP 5x5mm P0.5mm	Differential RF drivers feeding TFLN/MZM modulation chain.
U30-U35	HBM4 for NCE-A	HBM4-16GB SK_HBM4_16GB_12H	LightRail:BGA-1024_12x9mm	Six HBM4 stacks near U1.
U40-U45	HBM4 for NCE-B	HBM4-16GB SK_HBM4_16GB_12H	LightRail:BGA-1024_12x9mm	Six HBM4 stacks near U4.

Net / pin cues from native schematic:

- NCE symbols expose TFLN_TX_P0..P3 and TFLN_TX_N0..N3 pins.
- TFLN U3 exposes TX_P0..P3, TX_N0..N3, B-side TX inputs, OPT_TX0..OPT_TX3, VDD_RF, BIAS_V1 and MON_PD.
- The original A0 KiCad schematic PDF still remains in the package; this overview is provided for readability.